

Vertebral Column and Spinal Cord

Department Anatomy
Neuroanatomy (4)

Third Part

Dr. Osman Ahmed Osman

Lectuer Note:

➤ Introduction and Review of The vertebral column.

- The vertebral column.
- Vertebrae
- Gneral featuers of vertebrae.
- Curvatuers of vertiberal colum.
- Cervical Vertebrae.
- First and Second Cervical Vertebrae.
- The Thoracic Vertebrae.
- The lumber vertebrae.
- The Joints between vertebrae.
- Ligaments between vertebrae.
- The disc.
- Blood supply of vertebra and spinal cord.
- Nerve Supply of Vertebral Joints.
- Dermatomes innervated by posterior rami of spinal nerves
- Intervertebral foramen

➤ The Spinal Cord

- Introduction of Spinal Cord.
- Lumbar Puncture (Spinal Tap).
- Curvatuers of vertiberal colum deferate end of spinal cord between adult and infant and young child.
- Function of spinal cord.
- Spinal cord has:
 - External Features.
 - Internal Featuers.
- Spinal Meninges amd spaces.
- Spinal Segments.
- Internal Featuers.
- The gray matter of the spinal cord.
- The White matter of Spinal Cord
- Transverse sections of the spinal cord at different levels showing the arrangement of the gray matter and white matter.

➤ Important notes about the white mater of Spinal Cord.

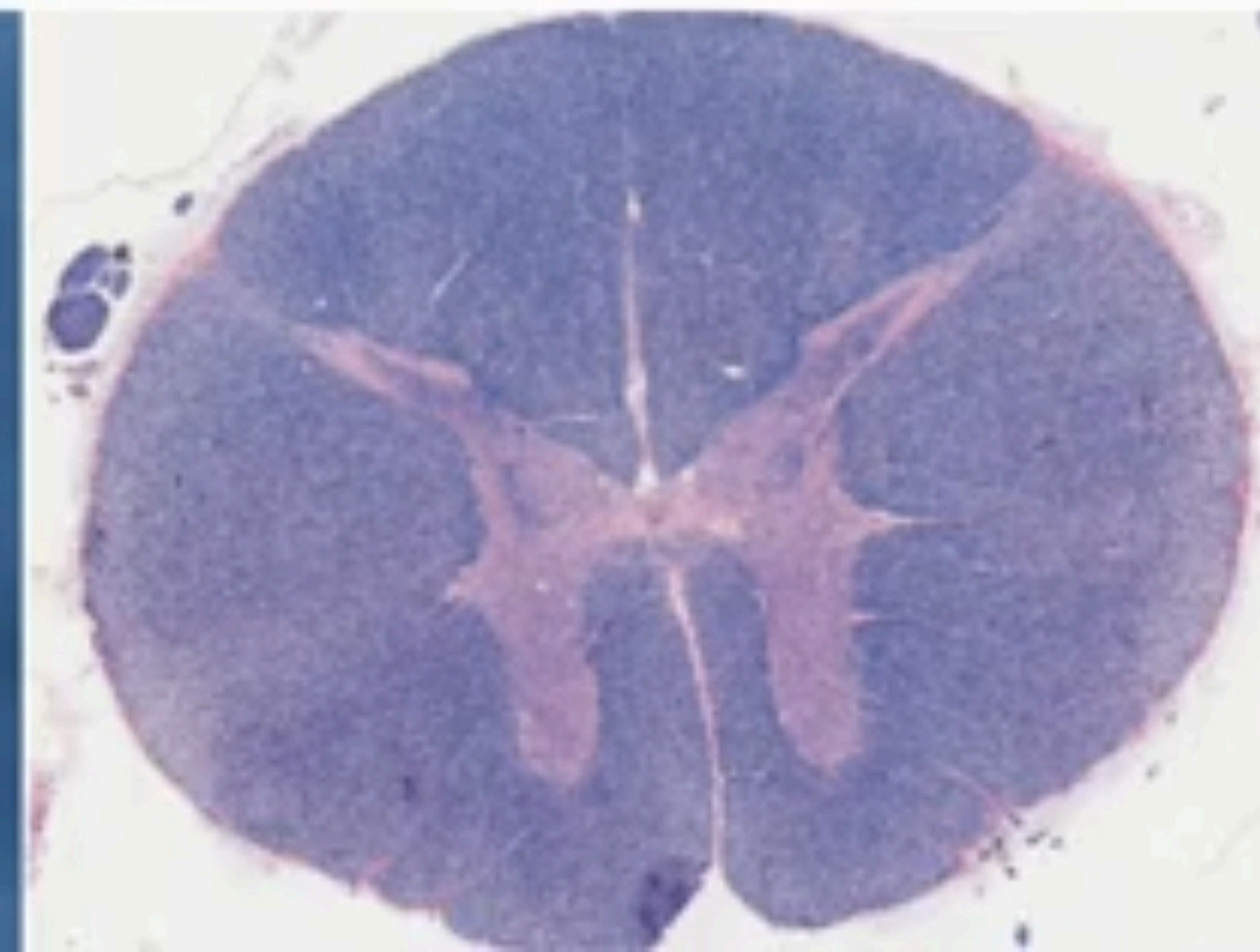
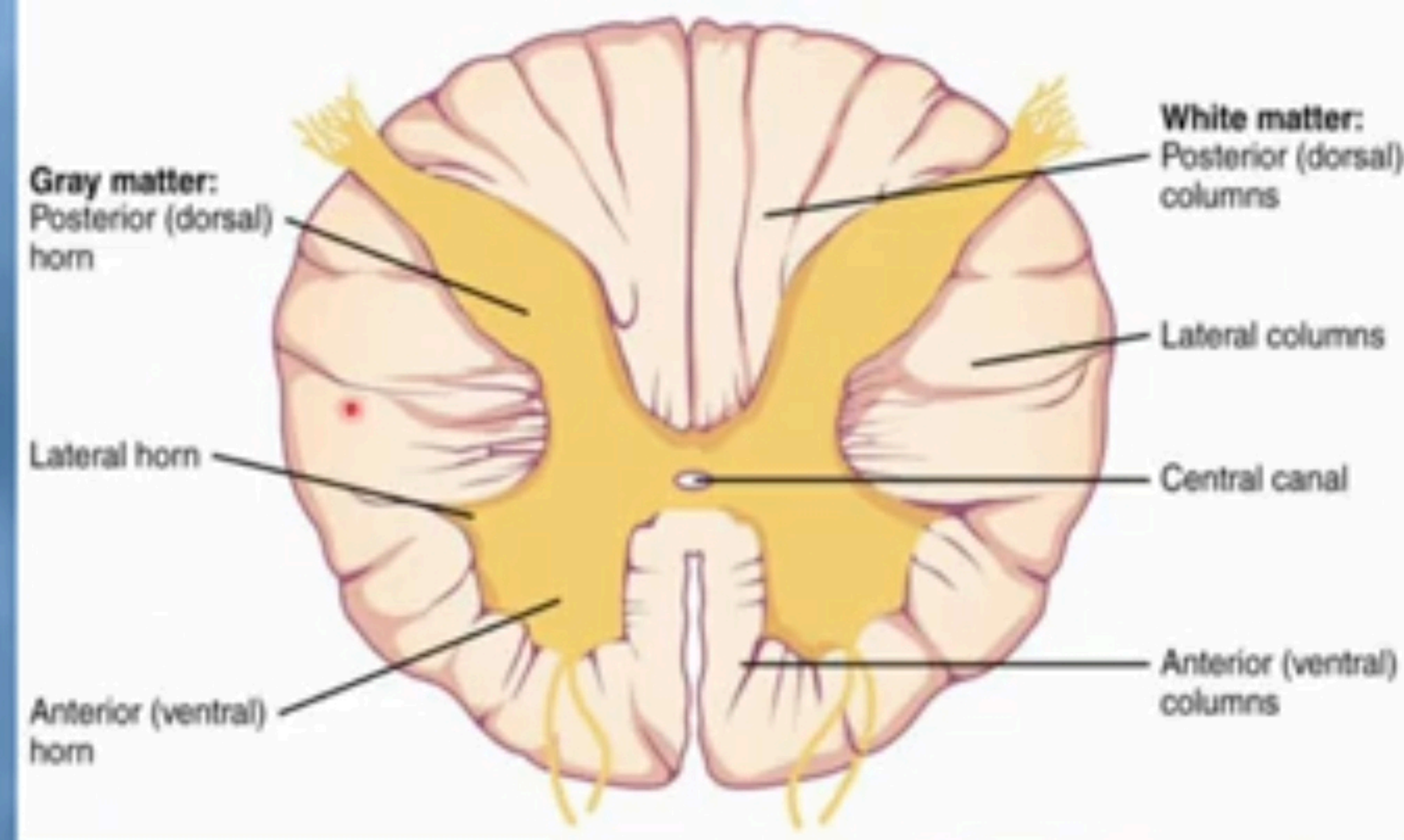
➤ Spinal Nerves.

- Transverse section of the spinal cord at the mid cervical level showing the general arrangement of the ascending tracts on the right and the descending tracts on the left.
- Functional components of spinal nerve fibers .
- Components and branches of spinal nerves.
- Arterial supply and venous drainage of spinal cord.
- Histological Structure of Nerve.
- Dermatomes.
- Myotomes.
- Main Descending and ascending Main pathway .

Internal Spinal Features

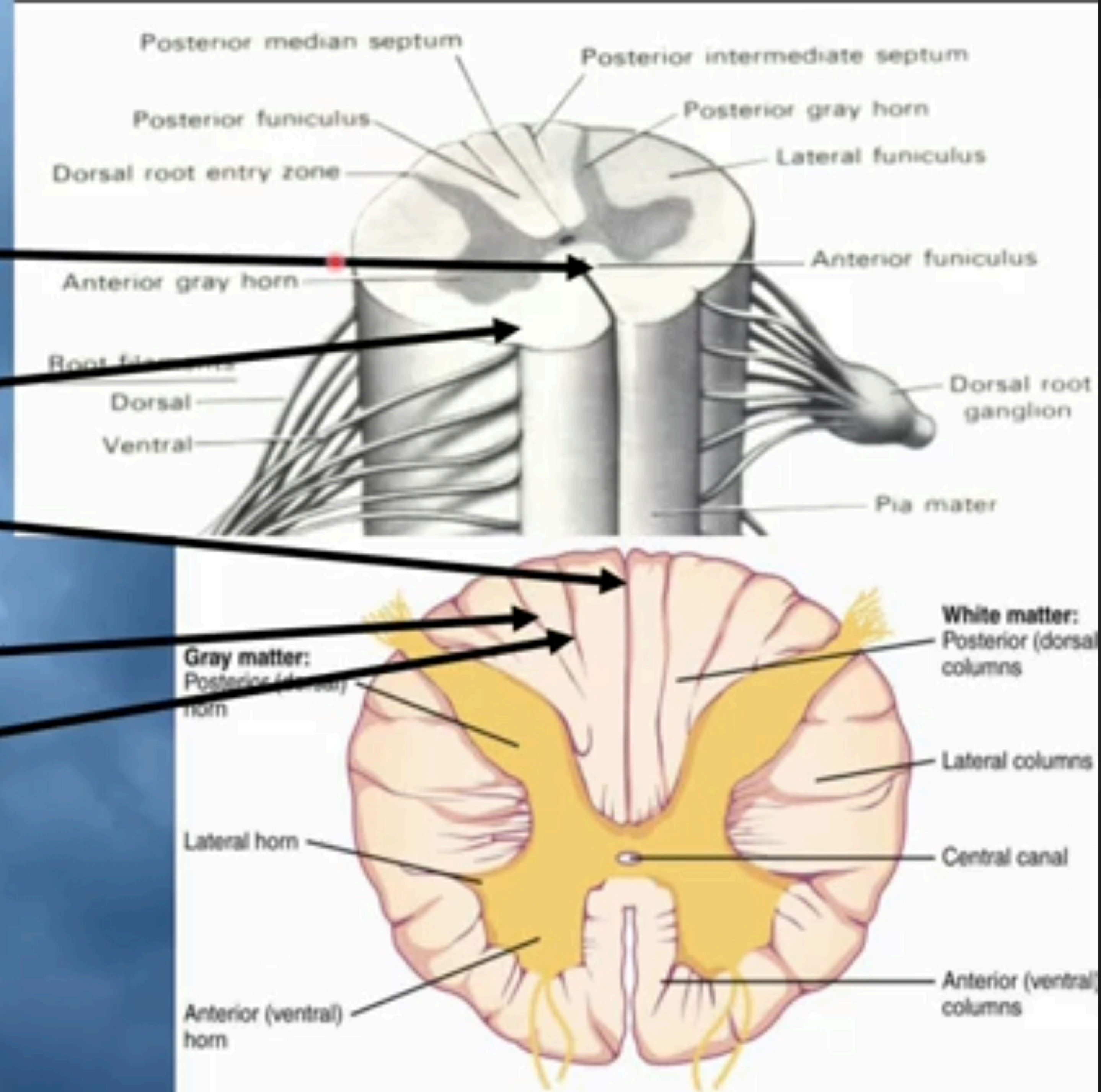
➤ The spinal cord is composed of:

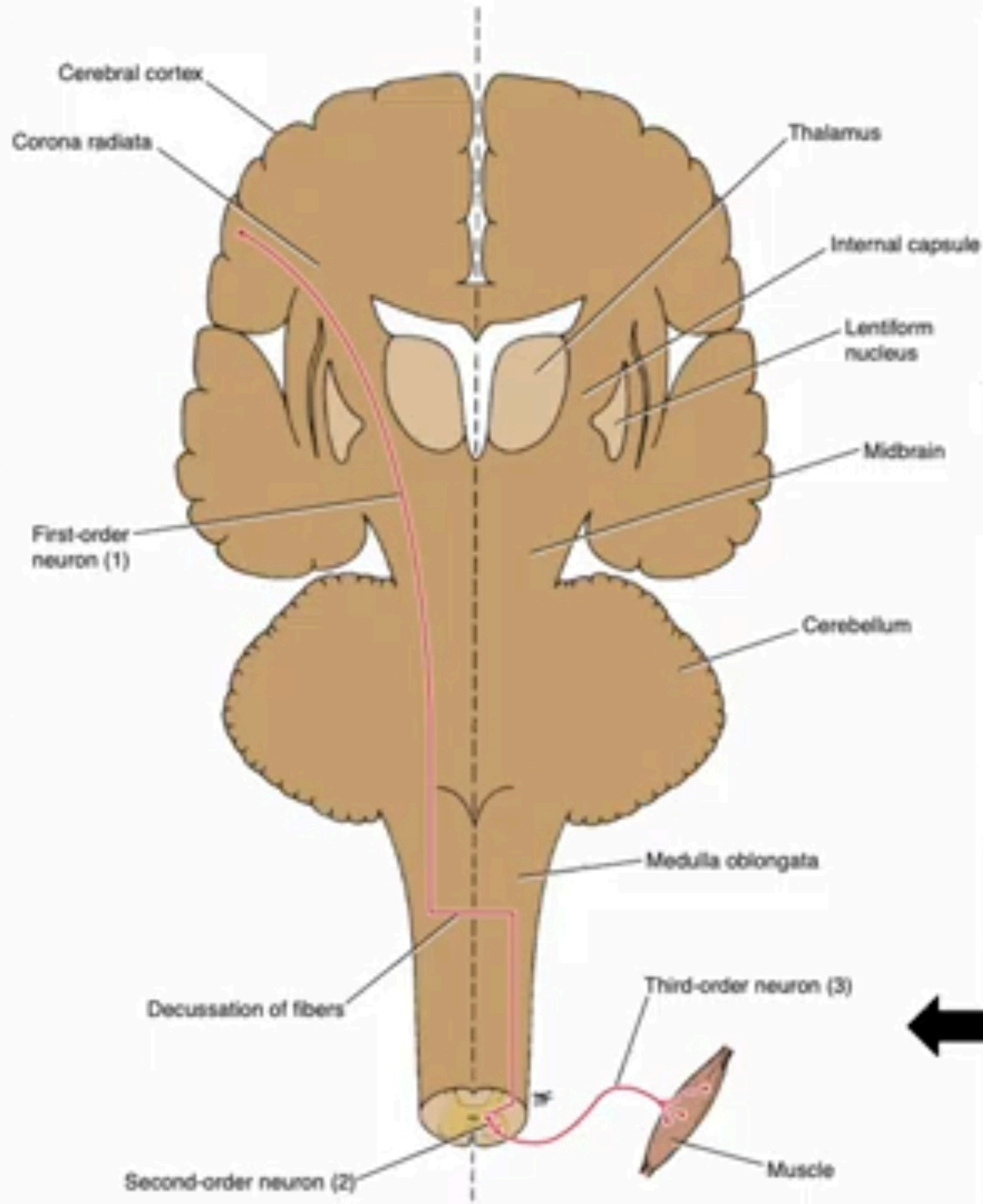
- 1) The inner core of gray matter.
- 2) The outer covering of white matter (which is surrounded the gray matter)



Internal Spinal Features

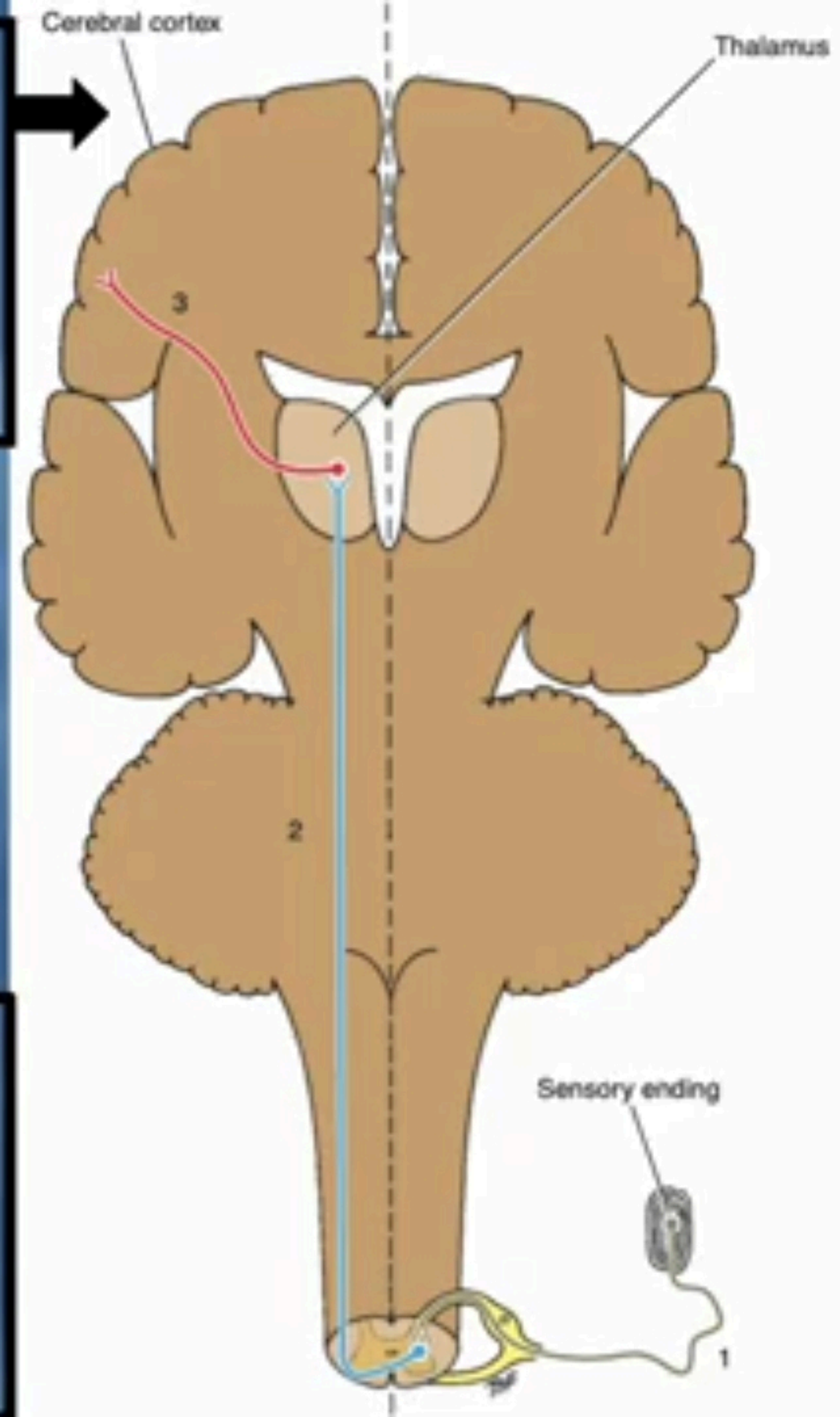
- Anterior median fissure in midline.
- Anteriolateral sulcus.
- Posterior median septum in midline.
- Posteriolateral sulcus.
- Posterior intermediate septum between the posteriolateral and the midline posteriorly.





Simplest form of the **ascending sensory pathway** from the sensory nerve ending to the cerebral cortex. Note the three neurons involved.

Simple form of the **descending motor pathway** from the cerebral cortex to the skeletal muscle. Note the three neurons involved.



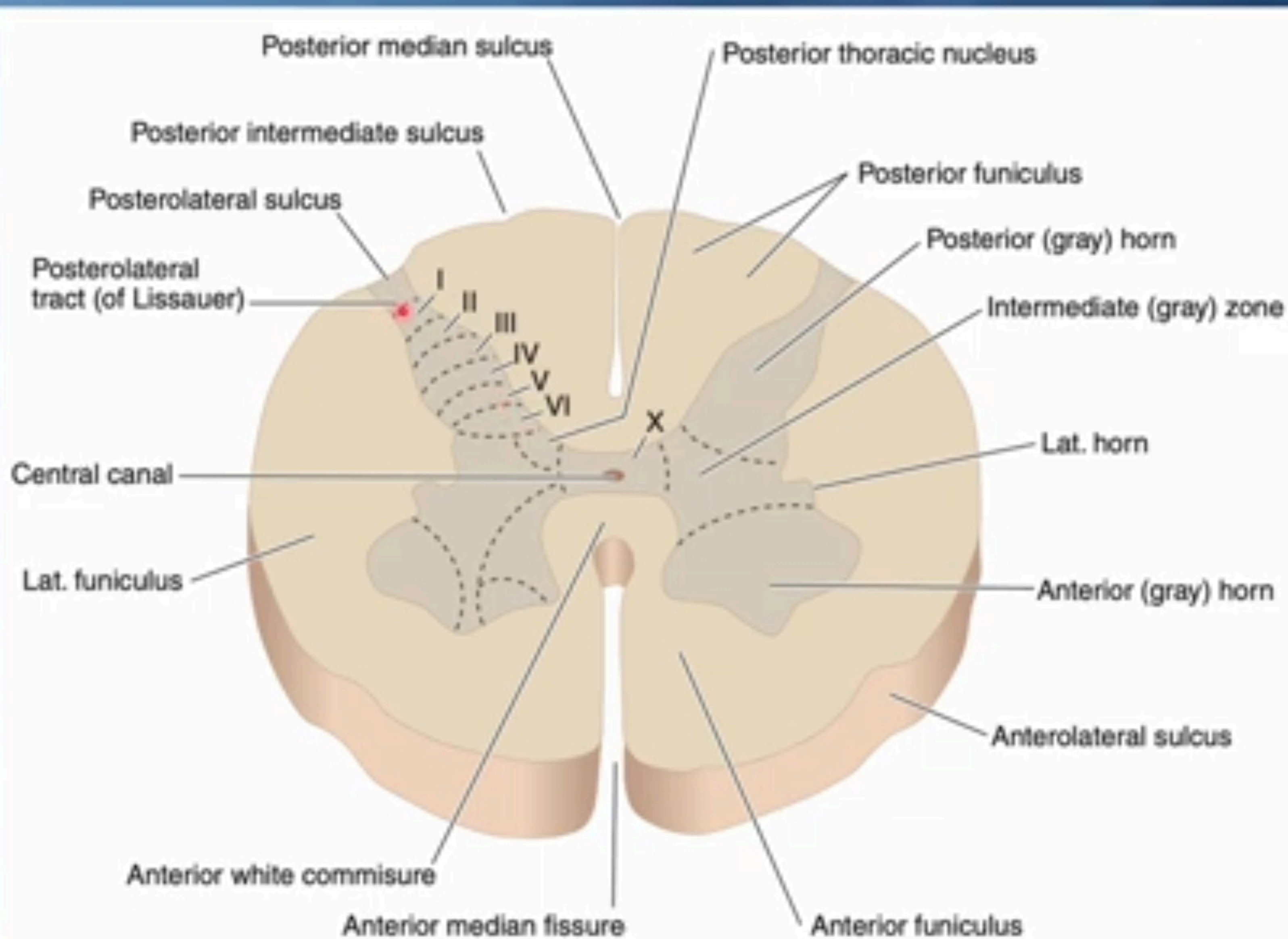
**The gray matter of the spinal cord consists of:
(a mixture of nerve cells and their processes, neuroglia, and blood vessels):**

1) Nerve Cell Groups in the Anterior Gray Columns.

2) Nerve Cell Groups in the Posterior Gray Columns.

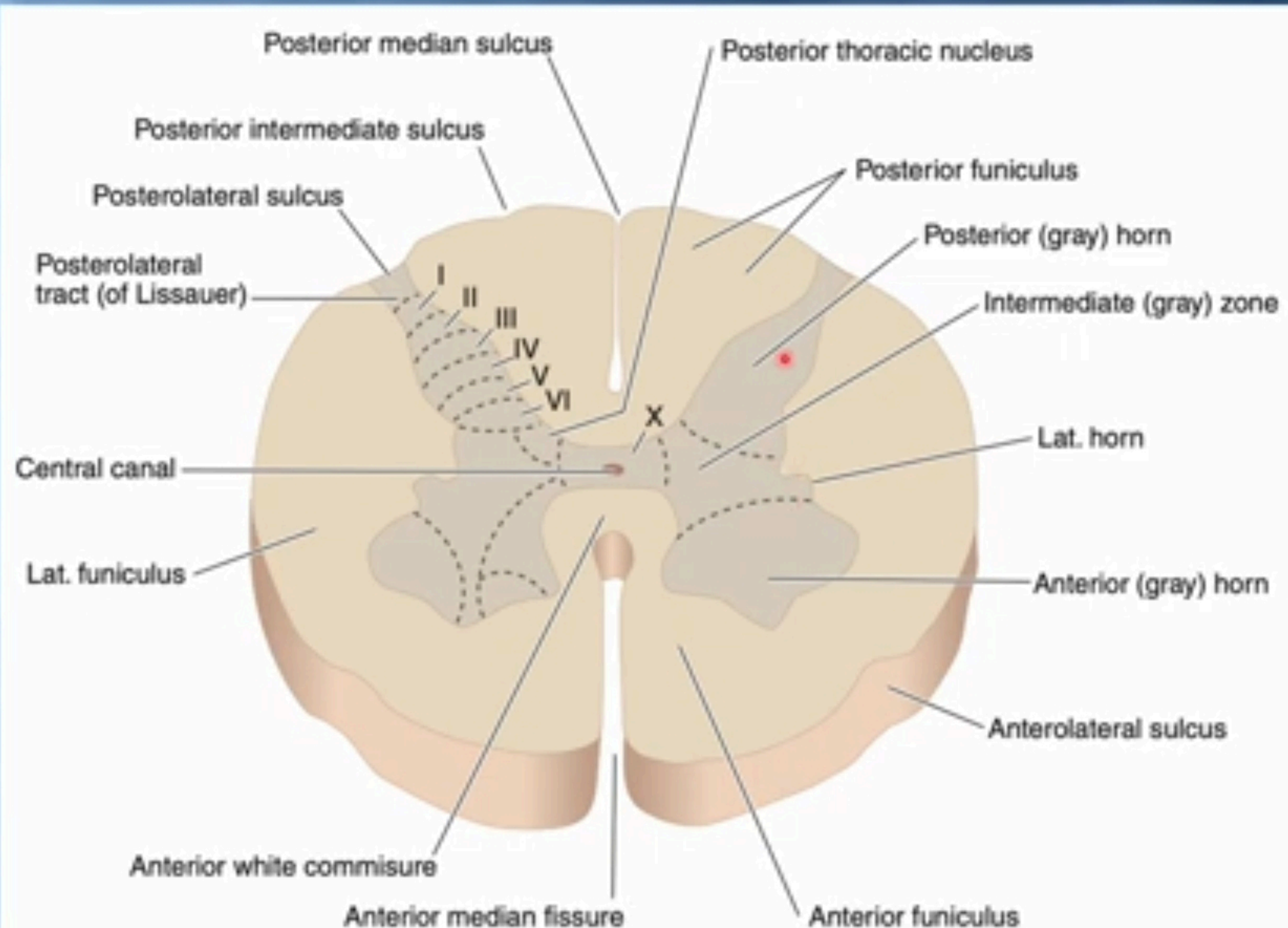
3) Nerve Cell Groups in the Lateral Gray Columns.

4) The Gray Commissure and Central Canal.



- It is found in the center of the spinal cord.
- It is **butterfly-** or **H-shaped** that varies according to spinal cord level.
- It contains a **central canal**.
- It divided into **cytoarchitectural areas** called **Rexed laminae**, expressed with **Roman numerals**.
- It divided into **three horns** or **cell columns** on each side.

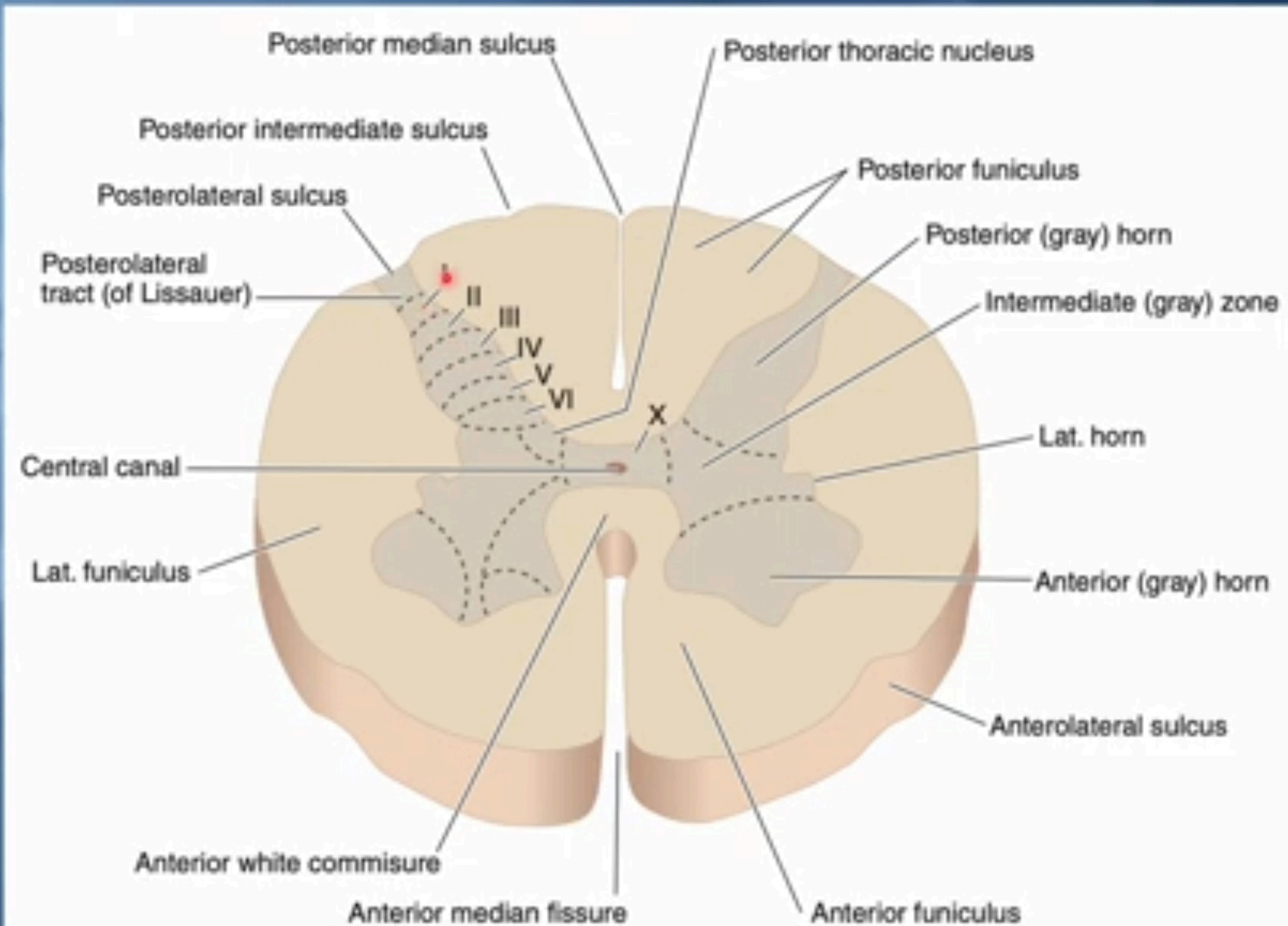
The gray matter of the spinal cord



1) Posterior horn (column)

- It receives and processes **sensory input.**
- There are **four nerve cell groups** of the posterior gray column:
 - ✓ Two that extend throughout the length of the cord and
 - Two that are restricted to the thoracic and lumbar segments.
- It includes the following nuclei:
 - (1) **Posteromarginal nucleus (Rexed lamina I)**
 - (2) **Substantia gelatinosa (Rexed lamina II)**
 - (3) **Nucleus proprius (Rexed laminae III and IV)**
 - (4) **Posterior thoracic nucleus (also known as Nucleus dorsalis of Clarke) (Rexed lamina VII)**
- ✓ **The visceral afferent nucleus.**

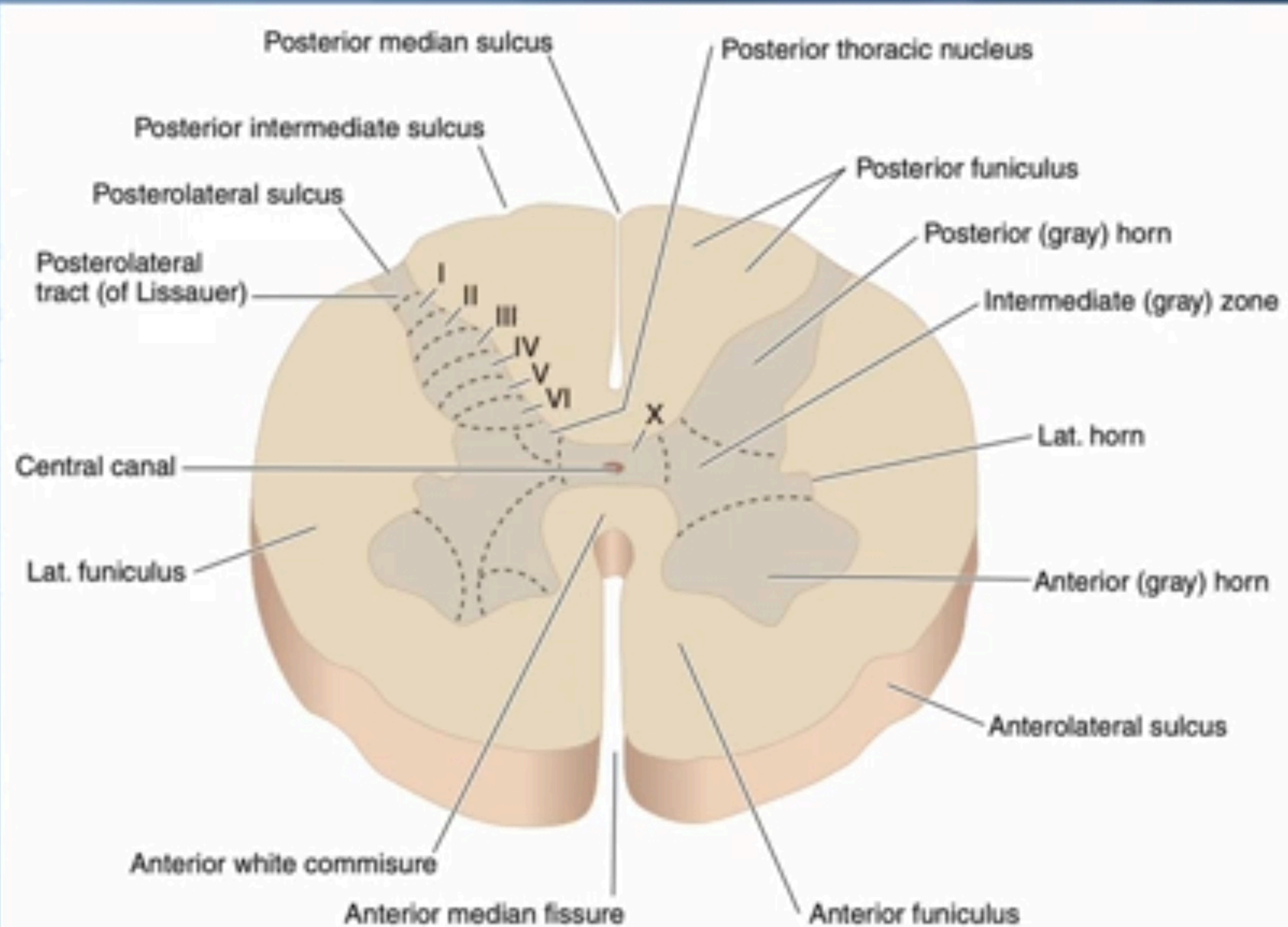
The gray matter of the spinal cord



2. Lateral horn (column) (Rexed lamina VII)

- It receives **viscerosensory** input.
- It found between the **posterior** and **anterior horns**. extends from **T1** to **L2**.
- It contains the **intermediolateral nucleus** (column), a visceromotor nucleus that extends from **T1** to **L2**.
- It contains **preganglionic sympathetic neurons (GVE)**.
- It contains, **at T1–T2**, the **cilio-spinal center** of Budge (sympathetic innervation of the eye).

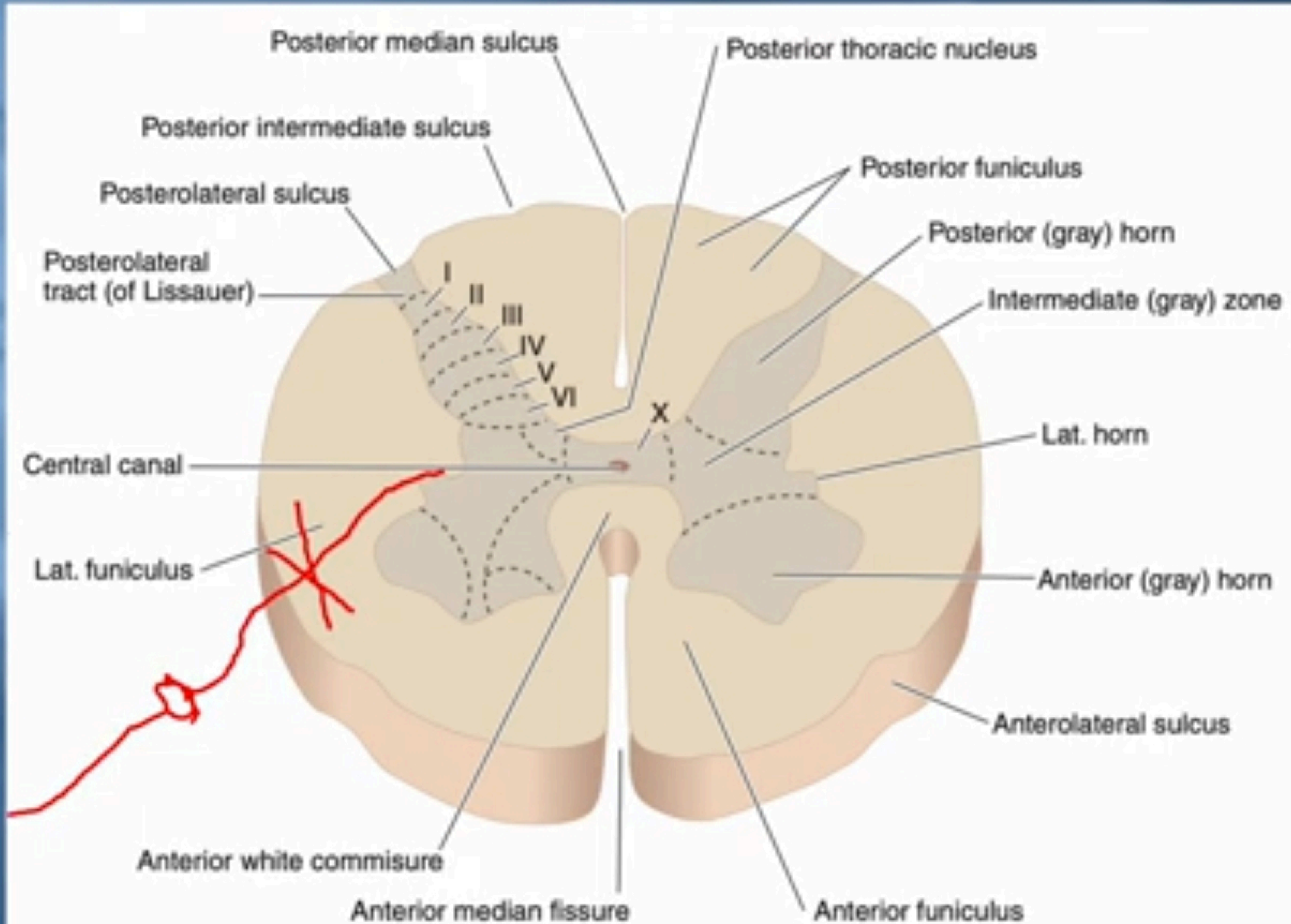
The gray matter of the spinal cord



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The gray matter of the spinal cord

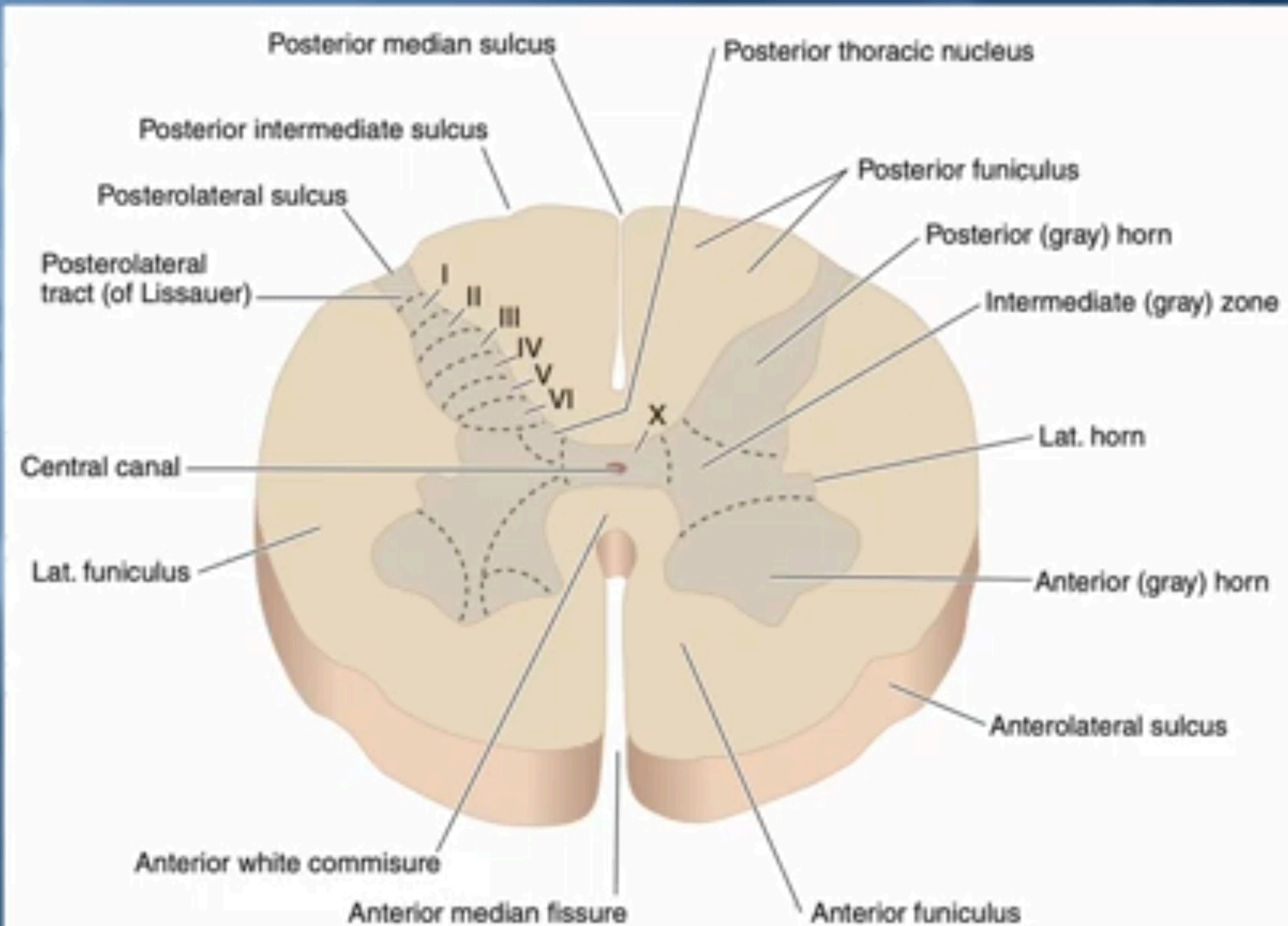


3. Anterior horn (column) (Rexed laminae VII, VIII, and IX)

- It contains predominantly motor nuclei.
- found at all levels.
- It includes the following nuclei:

- (1) Spinal border cells.
- (2) Sacral parasympathetic nucleus (Rexed lamina VII)
- (3) Somatic motor nuclei (Rexed lamina IX).
- (4) Spinal accessory nucleus (Rexed lamina IX)
- (5) Phrenic nucleus (Rexed lamina IX).

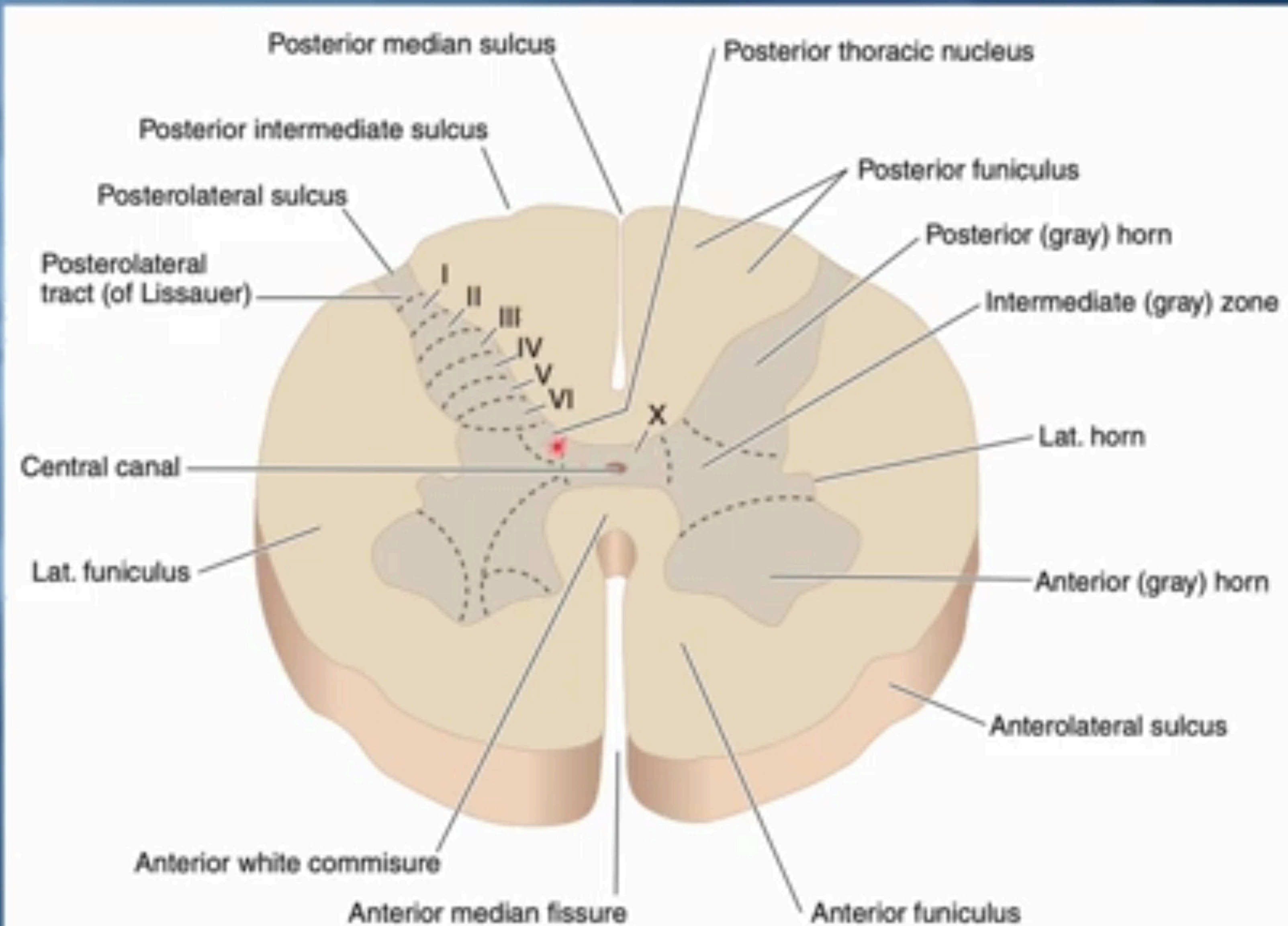
The gray matter of the spinal cord



4. The Gray Commissure and Central Canal

- **The Gray Commissure**
 - **The posterior gray commissure.**
 - **The anterior gray commissure.**
- **The Central Canal**
 - It is filled with cerebrospinal fluid and
 - It is lined with ciliated columnar epithelium, the ependyma.

The gray matter of the spinal cord



White matter of Spinal Cord

White matter:

- It consists of bundles of myelinated fibers that surround the central gray matter.
- It consists of ascending and descending fiber pathways called tracts.

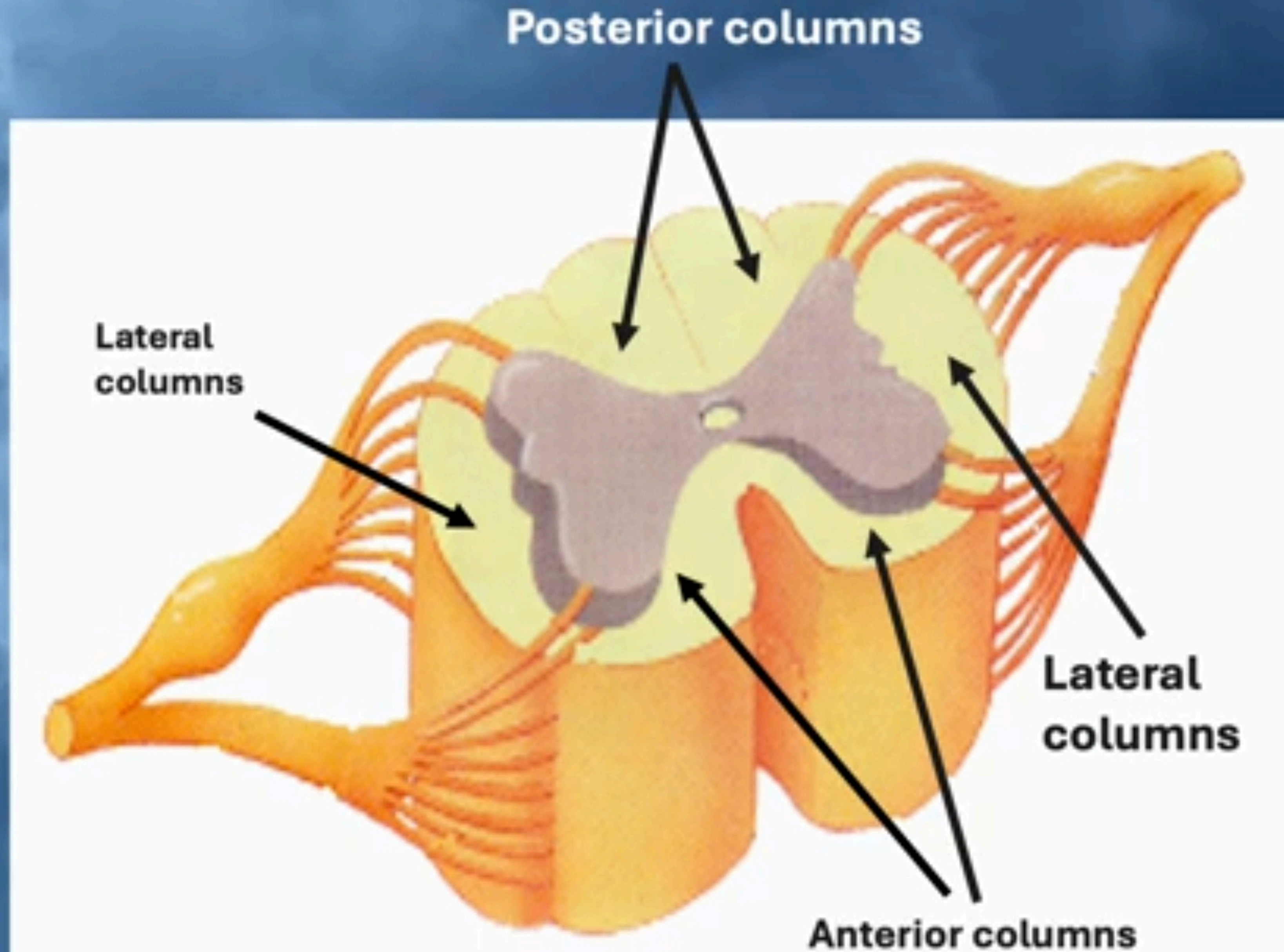
➤ It divided bilaterally by sulci into three major divisions.

1. Posterior funiculus (posterior column)

- it is subdivided above T6 into two fasciculi:
 - a. Fasciculus gracilis.
 - b. Fasciculus cuneatus.

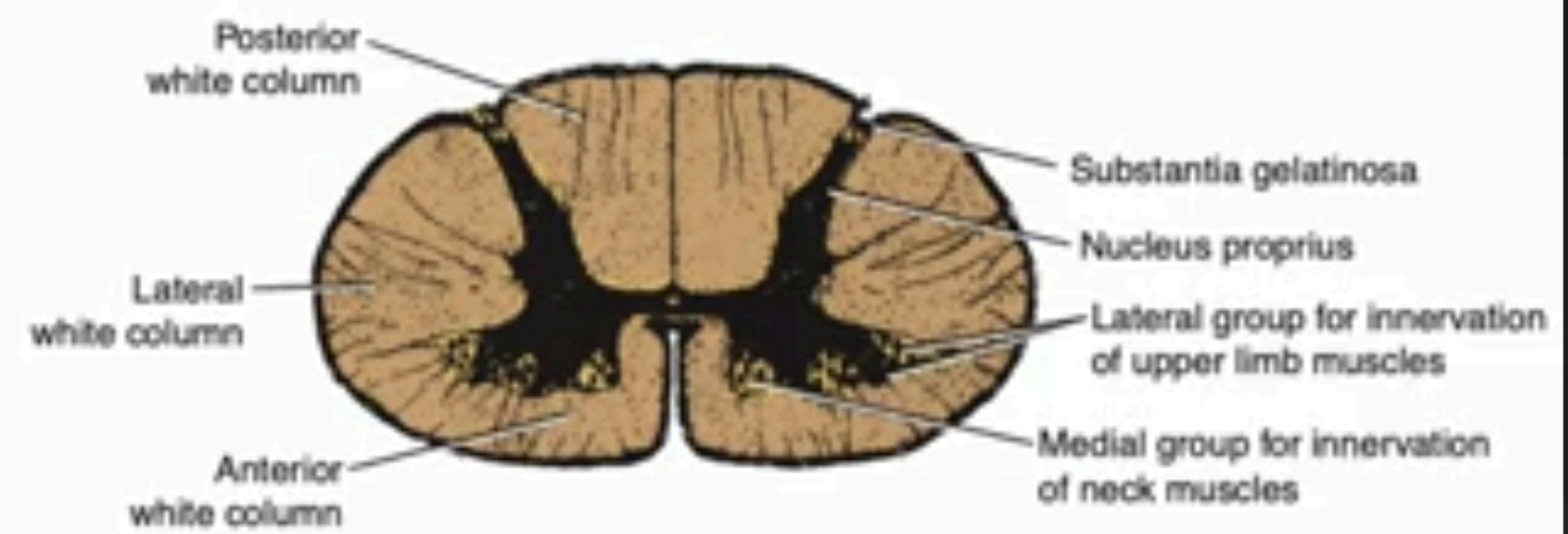
2. Lateral funiculus.

3. Anterior funiculus.

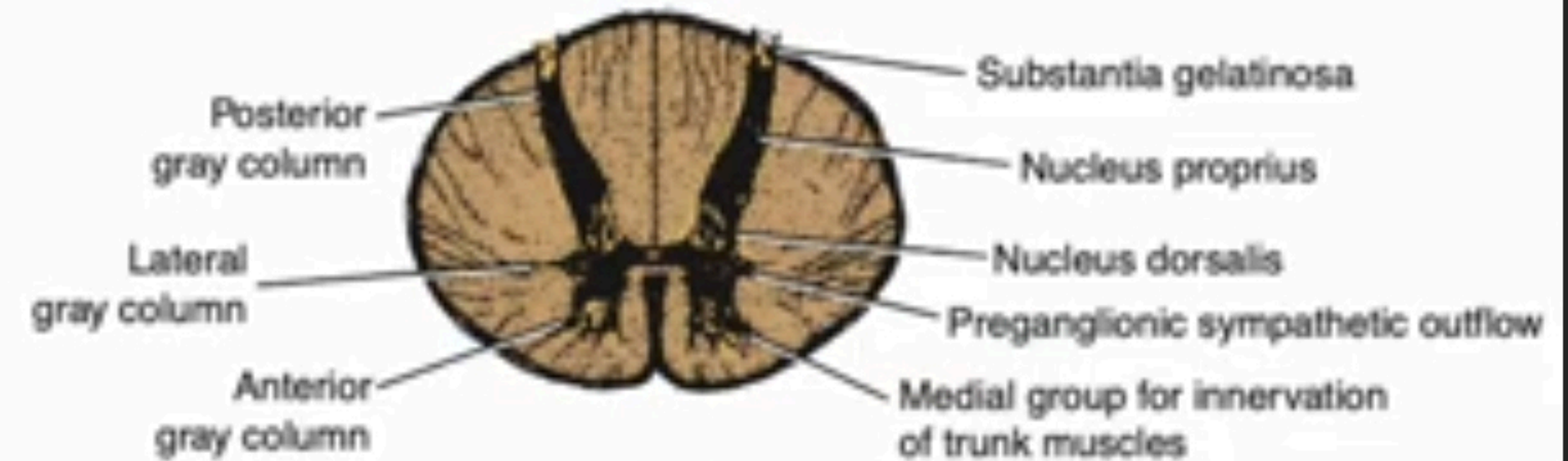


Transverse sections of the spinal cord at different levels showing the arrangement of the gray matter and white matter.

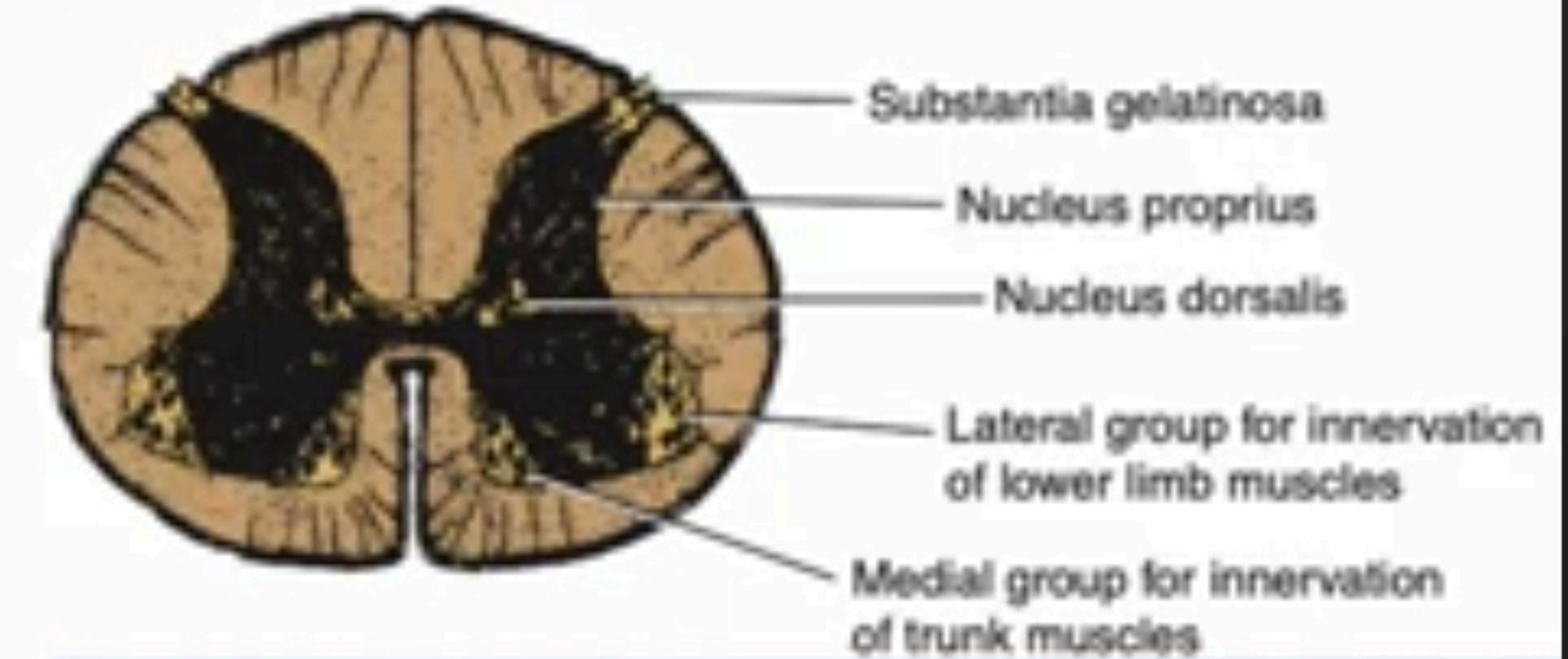
Sixth cervical segment



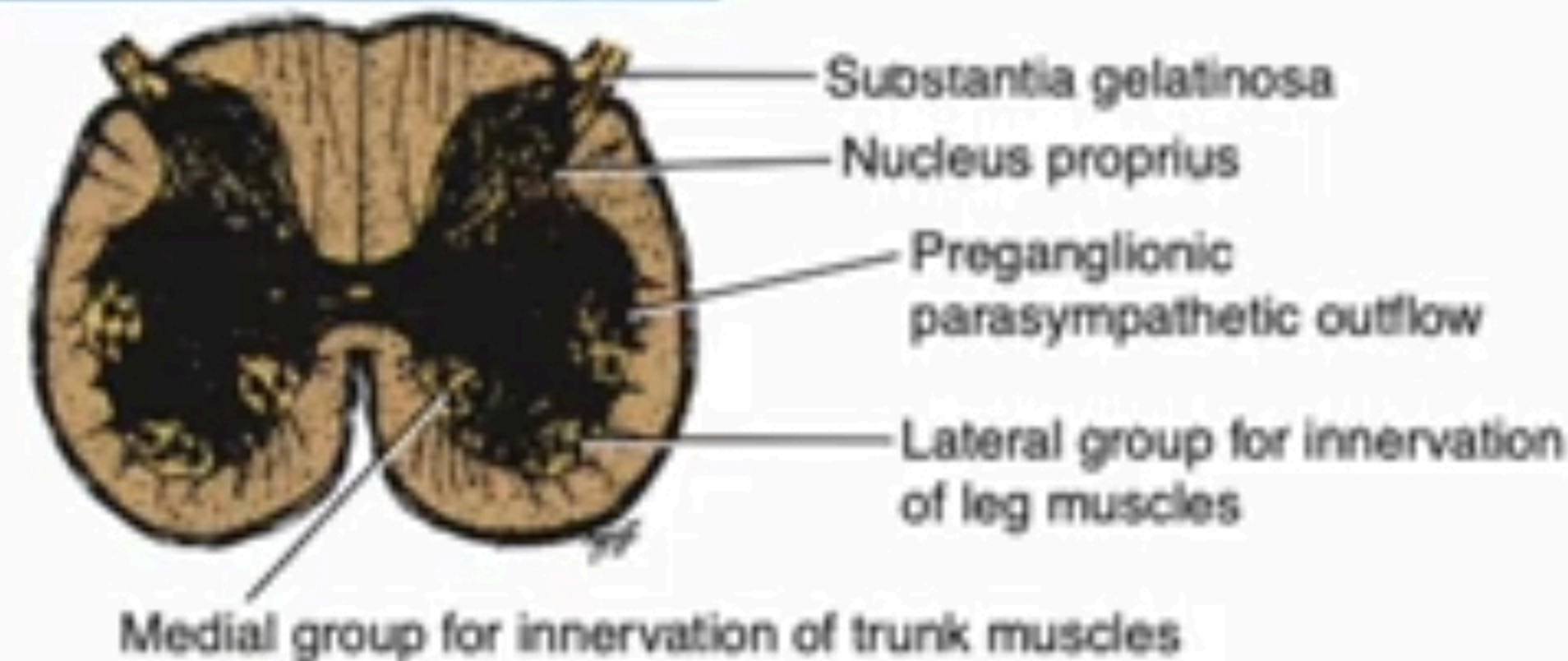
Sixth thoracic segment



Third lumbar segment



Third sacral segment



Comparison of Structural Details in Different Regions of the Spinal Cord

Region	Shape	White Matter	Gray Matter		
			Anterior Gray Column	Posterior Gray Column	Lateral Gray Column
Cervical	Oval	Fasciculus cuneatus and fasciculus gracilis present	Medial group of cells for neck muscles; central group of cells for accessory nucleus (C1-5) and phrenic nucleus (C3-5); lateral group of cells for upper limb muscles	Substantia gelatinosa present, continuous with Sp.N. of cranial nerve V at level C2; nucleus proprius present; nucleus dorsalis (Clarke's column) absent	Absent
Thoracic	Round	Fasciculus cuneatus (T1-6) and fasciculus gracilis present	Medial group of cells for trunk muscles	Substantia gelatinosa, nucleus proprius, nucleus dorsalis (Clarke's column), and visceral afferent nucleus present	Present; gives rise to preganglionic sympathetic fibers
Lumbar	Round to oval	Fasciculus cuneatus absent; fasciculus gracilis present	Medial group of cells for lower limb muscles; central group of cells for lumbosacral nerve	Substantia gelatinosa, nucleus proprius, nucleus dorsalis (Clarke's column) at L1-4, and visceral afferent nucleus present	Present (L1-2 [3]); gives rise to preganglionic sympathetic fibers
Sacral	Round	Small amount; fasciculus cuneatus absent; fasciculus gracilis present	Medial group of cells for lower limb and perineal muscles	Substantia gelatinosa and nucleus proprius present	Absent; group of cells present at S2-4, for parasympathetic outflow



Will be continue...